THE PULLBACK OF AN INJECTIVE MODULE ALONG A CONTINUOUS COCONTINUOUS FUNCTOR HAS NO HIGHER COHOMOLOGY ON OBJECTS OF THE SOURCE SITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the reduction to Cech cohomology, the identification of the two Cech complexes through cocontinuity and the fibre-product compatibility, and the final vanishing from injectivity.

Editorial note. The label lemma-pullback-injective-pre-limp names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement and proof are copied from the pinned chapter file; the three cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Cech-to-derived comparison for a covering. Same chapter, the lemma deducing the vanishing of cohomology from the vanishing of all higher Cech cohomology groups of coverings. Supplies the reduction from cohomology to Cech cohomology.
- The pullback along a morphism of topoi as the presheaf pullback. Sites, the lemma identifying g^{-1} with the presheaf pullback u^p for a continuous and cocontinuous functor. Supplies the identification of the Cech complexes used in the comparison.
- Injective modules have no higher Cech cohomology. Same chapter, the lemma stating that the higher Cech cohomology of an injective module vanishes. Supplies the vanishing of the Cech groups on the image covering.

Source reference. Original reference 03F9: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 1565–1565. Prerequisite pointer invoked at proof line(s) 9289; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 03FC: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 1816–1816. Prerequisite pointer invoked at proof line(s) 9306; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original lemma 00XR: sources/stacks-chow/tags.txt, source lines 1229–1229. Prerequisite pointer invoked at proof line(s) 9303; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $u : \mathcal{C} \to \mathcal{D}$ be a continuous and cocontinuous functor of sites. Let $g : Sh(\mathcal{C}) \to Sh(\mathcal{D})$ be the corresponding morphism of topoi. Let $\mathcal{O}_\mathcal{D}$ be a sheaf of rings and let $\mathcal{I}$ be an injective $\mathcal{O}_\mathcal{D}$-module. Then $H^p(U, g^{-1}\mathcal{I}) = 0$ for all $p > 0$ and $U \in \mathrm{Ob}(\mathcal{C})$.*

Proof. The vanishing of the lemma follows from Lemma 03F9 if we can prove vanishing of all higher Čech cohomology groups $\check{H}^p(\mathcal{U}, g^{-1}\mathcal{I})$ for any covering $\mathcal{U} = \{U_i \to U\}$ of $\mathcal{C}$. Since u is continuous, $u(\mathcal{U}) = \{u(U_i) \to u(U)\}$ is a covering of $\mathcal{D}$, and $u(U_{i_0} \times_U \ldots \times_U U_{i_n}) = u(U_{i_0}) \times_{u(U)} \ldots \times_{u(U)} u(U_{i_n})$. Thus we have

$$\check{H}^p(\mathcal{U}, g^{-1}\mathcal{I}) = \check{H}^p(u(\mathcal{U}), \mathcal{I})$$

because $g^{-1} = u^p$ by Sites, Lemma 00XR. Since $\mathcal{I}$ is an injective $\mathcal{O}_\mathcal{D}$-module these Čech cohomology groups vanish, see Lemma 03FC. □